SRAG: Structured Reconstruction-Augmented Generation

Abstract

SRAG introduces a reconstruction-first paradigm for document intelligence. Unlike traditional Retrieval-Augmented Generation systems that rely on flat text retrieval, SRAG transforms unstructured and multimodal documents into structured knowledge representations. These representations are dynamically reconstructed based on query intent, enabling improved reasoning over tables, images, and complex layouts. The architecture integrates multimodal parsing, graph-based linking, and a Semantic Reconstruction Engine (SRE) to enhance accuracy and reduce hallucinations.

Architecture Diagram (Conceptual)

Document → Parsing Layer → Structuring Layer → Knowledge Graph → Multi-Index Storage → Query Planner → Reconstruction Engine (SRE) → Generation Layer

Methodology

1. Multimodal Parsing: Extract text, tables, and images from documents
2. Structuring: Convert extracted data into semantic units
3. Graph Construction: Link units via relationships
4. Hybrid Retrieval: Combine vector and keyword search
5. Query Planning: Decompose and classify queries
6. Reconstruction: Rebuild structured knowledge before answering
7. Generation: Use LLM to produce final response

Experiments

1. Compare SRAG vs Traditional RAG on scanned PDFs
2. Evaluate table-based query accuracy
3. Measure hallucination reduction
4. Test retrieval precision using hybrid search
5. Benchmark response quality on multimodal queries

Semantic Reconstruction Engine (SRE)

The SRE is the core component of SRAG responsible for transforming noisy inputs into structured knowledge.

1. Input: OCR outputs, parsed elements, graph nodes
2. Processing: Normalize text, reconstruct tables, resolve relationships
3. Output: Clean structured representation (JSON-like schema)
4. Capabilities: Error correction, schema inference, context alignment

SRE Internal Architecture Diagram

OCR/Text + Layout + Graph Nodes → Noise Cleaner → Structure Detector → Table Rebuilder → Relationship Resolver → Schema Normalizer → Confidence Scorer → Structured Output

SRE Workflow (Step-by-Step)

1. Step 1: Input Aggregation – Collect OCR text, layout data, and graph-linked nodes
2. Step 2: Noise Cleaning – Remove OCR errors, normalize tokens
3. Step 3: Structure Detection – Identify if data is paragraph, table, or mixed
4. Step 4: Table Reconstruction – Rebuild rows, columns, and headers
5. Step 5: Relationship Resolution – Link entities using graph context
6. Step 6: Schema Normalization – Convert into structured JSON-like format
7. Step 7: Confidence Scoring – Assign reliability score to reconstructed data
8. Step 8: Output Delivery – Send structured data to generation module

SRE Design Principles

1. Reconstruction-first approach
2. Modality-aware processing
3. Graph-assisted reasoning
4. Error-tolerant design
5. Explainable structured outputs

Conclusion

SRAG represents a shift from retrieval-based systems to reconstruction-driven intelligence. By integrating structure, multimodality, and adaptive reasoning, it enables more accurate and interpretable document understanding.